Solar Power Optimizer

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Goals

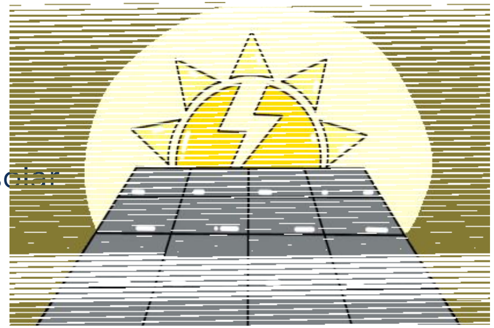
A predictive model to initiate maintenance activity on large solar farms is essential to reduce power interruption while controlling the cost involved in performing frequent repair activity. The project attempts to build such a predictive model based on a correlation analysis of data available from solar plants.

Introduction

Solar panels are electronic devices that absorb sunlight and convert it directly into usable electricity.

Typical Parameters

- **Solar Irradiance**: A direct measure of light energy hitting the panels. The higher the sunlight, the more electricity that can be generated.
- **Ambient temperature:** The temperature of the air around the solar panels.
- **Module temperature:** The actual temperature of the solar panel itself.
- **Panel Angle:** The angle at which the solar panel is positioned toward the sun.
- **Soiling index**: A measure of how much dirt or debris has been built up upon the solar panel.

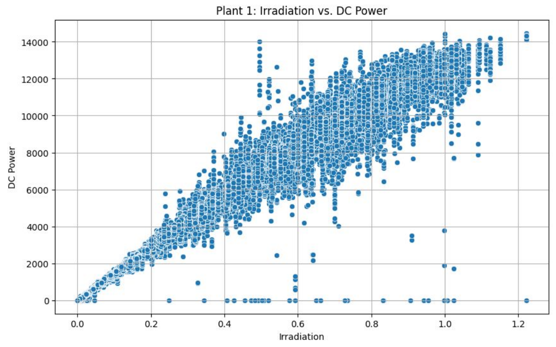


Methodology

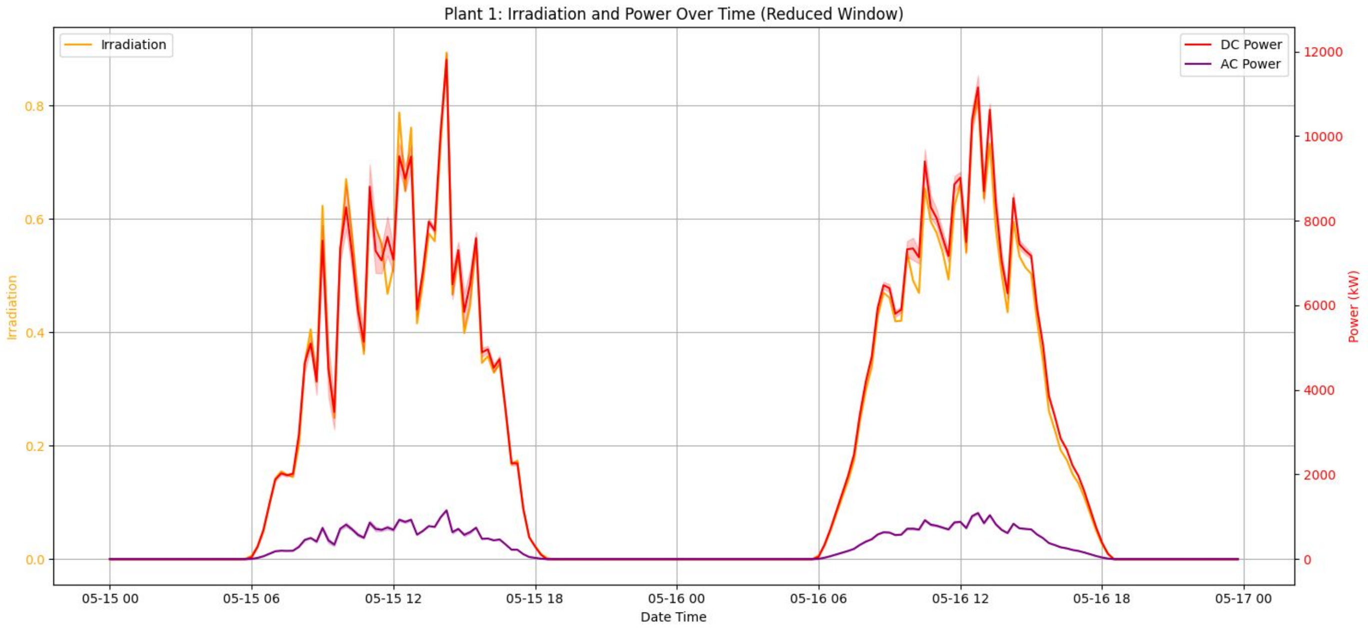
Data Source: https://www.kaggle.com/datasets/anikannal/solar-power-generation-data

- Derive the correlation between different variables that affect power generation.
- **Visualize the time series data** to identify the relationship between variables
- **Formal method. Used the correlation function in python** (shown below). This computes Pearson correlation coefficient that quantifies the strength and direction of a linear relationship between two variables. The values lie between -1 and 1, where negative values indicate inverse relationship, 0 indicates no correlation and positive values indicate direct correlation.

```
correlation_matrix = df.corr(numeric_only=True)
```



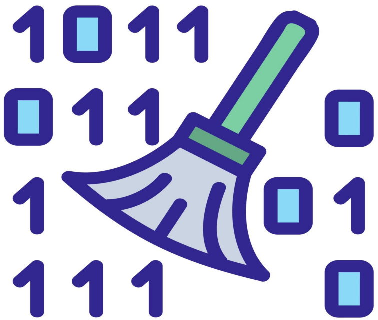
Correlation



The DC power is positively correlated with solar irradiation. As irradiation increases, DC power generation increases.

Formal: Correlation between Irradiation and DC Power for Plant 1: **0.9894**

Cleaning the data



- The sensor data is cleansed using linear interpolation when possible.
- A missing value is replaced with a value derived by drawing a straight line between the closest data points

```
df[numeric_cols] = df[numeric_cols].interpolate(method=’linear’, limit_direction=’both’)
```

Building the Model

1. Build a model using linear regression - predict a target value based on input by fitting a straight line through the data points in the equation below
   a. $Y = C0 + C1*X1 + C2*X2 + \dots + E$
2. Inputs (xi): Solar irradiation, ambient temperature
3. Output (y): Expected DC Power output (ideal DC power)
4. Model is trained using the downloaded data.

Maintenance Alert Logic

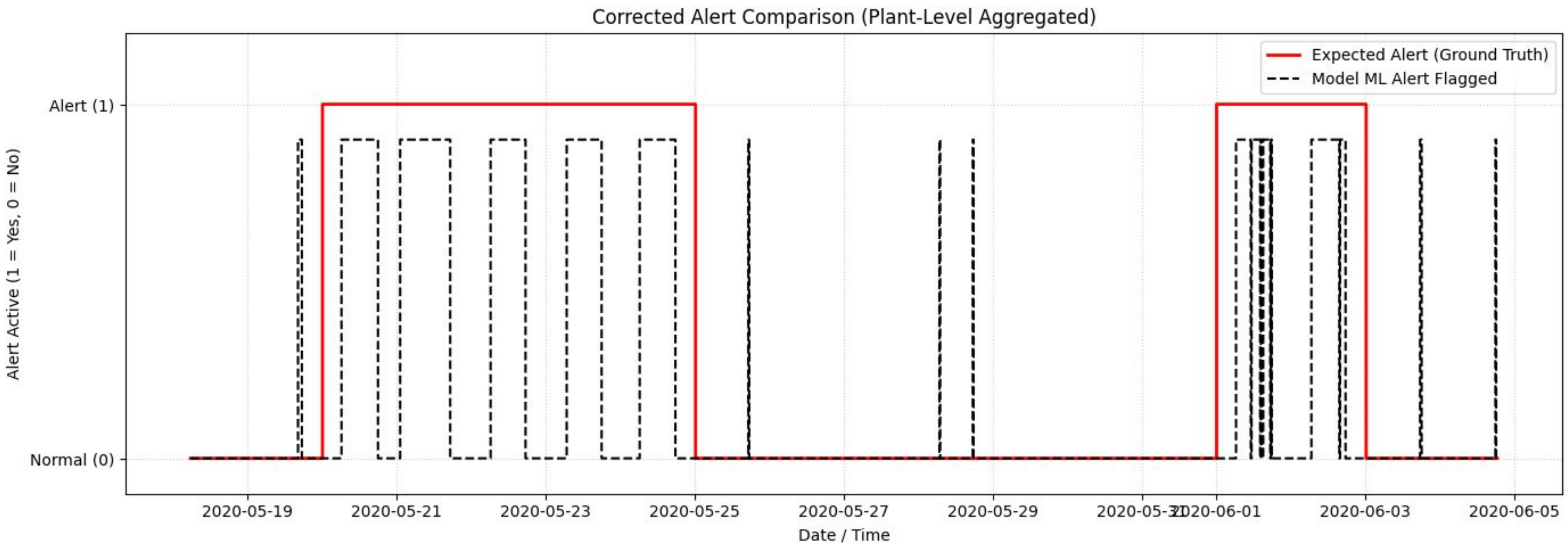
- Compute ideal power from the model
- Compute the ratio to actual power
- Alert if the ratio exceeds a threshold

```
# Rule 1: DC Power significantly lower than Ideal DC Power Only apply this rule when IDEAL_DC_POWER is positive (i.e., when there is irradiation)

condition_dc_power_drop = (df_alerts['IDEAL_DC_POWER'] > 0) &  (df_alerts['DC_POWER'] < ideal_dc_power_threshold * df_alerts['IDEAL_DC_POWER'])

df_alerts.loc[condition_dc_power_drop, 'ALERT'] = True
```

Evaluation Methodology

1. Inject artificial performance drops (both gradual and sudden) to simulate soiling, weather change etc.
2. Record the timestamps when the performance drops were injected
3. Generate the alerts using the model
4. Compare the actual alerts with the expected ones

Evaluation Results



A few false positives are observed (the black lines that are outside the red ones.

Future Work

- Reduce False positives
  - Learnings: When solar radiation is low (in the morning) the power output reduction is non-linear. Added a higher threshold to reduce false positives
- Tune the model